AI 101 for Mechanical Engineers · Module 10

Underfitting, overfitting and regularization

Lesson 10.2

What you will learn

> Distinguish underfitting from overfitting, interpret training and validation errors, explain regularization and early stopping, and check data and features before changing model complexity.

Lesson overview

Module 10 · Lesson 2

Underfitting, overfitting and regularization

AI 101 for Mechanical Engineers

1 Read training and validation errors

Underfitting



Misses a relationship the task needs. **Clue:** high error on both datasets.

Overfitting



Fits training-specific detail that does not transfer. **Clue:** training error much lower than validation error.

An error pattern is a clue, not proof. Before changing complexity, check the baseline, measurement noise, target labels and missing features.

2 Regularization controls fitting

Model fitting → Penalties or constraints → A restricted fit

For example, penalize large coefficients or limit model structure.

Too little flexibility can miss useful relationships (bias).	**Too much flexibility** can fit training-specific variation (variance).

Choose the controls using validation evidence. Here, bias means prediction error, not fairness.

L1 regularization (e.g. Lasso) can drive some coefficients to zero. A zero coefficient does not prove a sensor is physically irrelevant.

3 Early stopping uses validation



Training loss can keep falling while validation loss rises. Select the checkpoint within development.

Do not use the final test to select the epoch.

Key takeaway

Lower training error alone does not show better performance on relevant unseen data.

A selected checkpoint is not guaranteed to be best on every future example.

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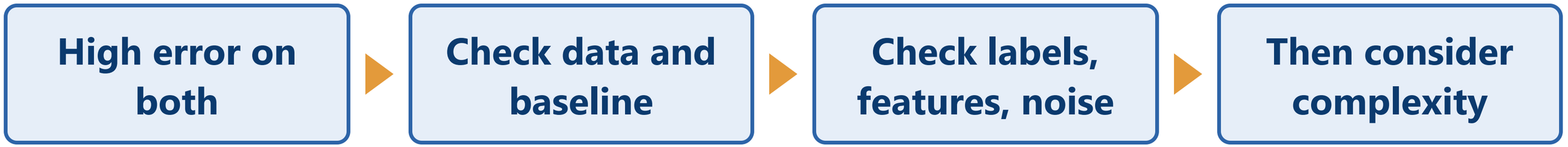
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Underfitting misses the relationship the task needs

- The model does not adequately represent the needed relationship
- Example: a straight line fitted to a strongly nonlinear response
- High training and validation errors can suggest underfitting
- Possible issues: low capacity, weak features, inadequate training

> Judge errors relative to a useful baseline, with sufficiently reliable data.

High error on both sets: check data first



- Noise, incorrect labels or missing features can also limit prediction
- A more flexible model gets the same incomplete inputs
- Inadequate representation and inadequate evidence need different investigations

Overfitting captures training-specific variation

- Fits training-specific variation; performs poorly on relevant unseen examples [4]
- Example: an irregular curve following incidental noise
- Deep decision trees can increase this risk
- Common sign: training error substantially below validation error

> Some difference is normal; no universal numerical gap proves overfitting.

Read training and validation errors together

Both errors high

- Investigate relationship, data and baseline

Low training, substantially poorer validation

- Investigate training-specific fitting

Underfitting and overfitting describe learning behaviour, not conclusions from a single score.

Regularization discourages certain complex fits

- Uses penalties or constraints on selected kinds of complexity
- Example: a penalty on large fitted coefficients
- Restricting too strongly can miss relevant relationships
- Bias and variance here concern prediction error, not fairness

> Schematic only: no penalty equation or universal setting is given. Use validation evidence.

Flexibility involves the bias-variance trade-off

More flexibility can

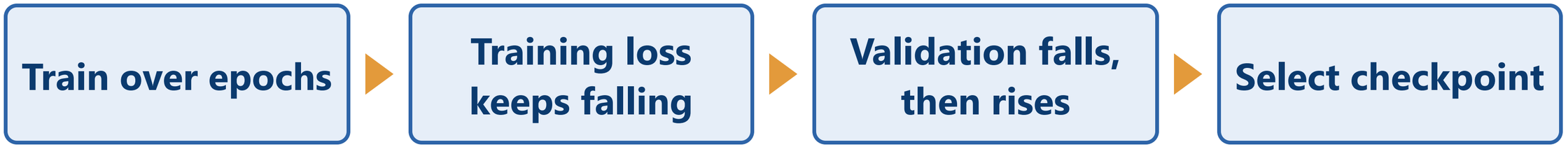
- Reduce systematic errors from overly simple fits

But can also

- Increase sensitivity to particular training examples [55]

Choose complexity with validation; do not assume lower training error means better unseen performance.

Early stopping selects a checkpoint on validation



- Checkpoint chosen using validation performance [56]
- The minimum does not guarantee optimal future performance
- The final-test score must not select the epoch

L1 can zero some coefficients; zero does not prove irrelevance

- L1 penalizes absolute coefficient sizes; can drive some to zero [57]
- This can produce a sparse linear model
- A zero coefficient does not prove a sensor irrelevant
- Lasso in scikit-learn: fit only within development data

> Correlated inputs, scaling and the training sample affect which coefficients remain.

CHECK YOUR UNDERSTANDING

Why does high error on both training and validation sets not always prove underfitting?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Why does high error on both training and validation sets not always prove underfitting?

> Noise, incorrect labels and missing features can also produce high errors. Investigate the data and compare with a useful baseline before concluding capacity is insufficient.

CHECK YOUR UNDERSTANDING

Why should an early-stopping checkpoint be chosen using validation rather than final-test results?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Why should an early-stopping checkpoint be chosen using validation rather than final-test results?

> Checkpoint selection is a development choice. Using final-test results would compromise the independent evaluation of the locked procedure.

Key takeaways

- Underfitting misses relationships; overfitting captures training-specific variation
- Read both errors with data quality and baseline evidence
- Regularization controls selected complex fits via penalties or constraints
- Early stopping selects a checkpoint in development; testing stays separate

References

[4] I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning. Cambridge, MA, USA: MIT Press, 2016.

[55] S. Geman, E. Bienenstock, and R. Doursat, “Neural networks and the bias/variance dilemma,” Neural Comput., vol. 4, no. 1, pp. 1–58, 1992. doi: 10.1162/neco.1992.4.1.1

[56] L. Prechelt, “Early stopping—But when?,” in Neural Networks: Tricks of the Trade (Lecture Notes in Computer Science, vol. 1524), G. B. Orr and K.-R. Müller, Eds. Berlin, Germany: Springer, 1998, pp. 55–69. doi: 10.1007/3-540-49430-8_3

[57] R. Tibshirani, “Regression shrinkage and selection via the lasso,” J. Roy. Statist. Soc. B, vol. 58, no. 1, pp. 267–288, 1996. doi: 10.1111/j.2517-6161.1996.tb02080.x

Numbers match the course reference list. The written lesson is the authoritative text.